

Martingale Matching

A novel approach to diffusion

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SDE Background

Definition (Filtration)

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let T be an index set. For each $i \in T$, let $\mathcal{F}_i$ be a sub- σ -algebra of $\mathcal{F}$. Then $\{\mathcal{F}_i\}_{i \in T}$ is a filtration if $\mathcal{F}_k \subset \mathcal{F}_l$ for each $k \leq l$. Then $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$ is called a filtered probability space.

Definition (Martingale)

A stochastic process $X : T \times \Omega \rightarrow \mathbb{R}$ on probability space $(\Omega, \mathcal{F}, P)$ is a martingale w.r.t. the filtration $\{\mathcal{F}\}$ (of $(\Omega, \mathcal{F}, P)$) if

- 1 X is adapted to the filtration. That is, X_t is $\mathcal{F}_t$ measurable for each $t \in T$.
- 2 for each t , $\mathbb{E}[X_t] < \infty$
- 3 for all s and t with $s < t$, $\mathbb{E}[X_t | \mathcal{F}_s] = X_s$.

Definition (Brownian motion)

Brownian motion is a stochastic process $\{B_t\}$ such that

- $B_0 = 0$
- for all k , and all $0 < t_1 < \dots < t_k$, the random variables

$$B_{t_k} - B_{t_{k-1}} \sim N(0, t_k - t_{k-1})$$

are independent

- B_t is almost surely continuous in t

Definition (Itô integral)

Let $f \in \nu(S, T)$. Then the Itô integral of f (from S to T) is defined by

$$\int_S^T f(t, \omega) dB_t(\omega) = \lim_{n \rightarrow \infty} \int_S^T \phi_n(t, \omega) dB_t(\omega) \quad (1)$$

where $\{\phi_n\}$ is a sequence of elementary functions such that

$$\mathbb{E} \left[\int_S^T (f(t, \omega) - \phi_n(t, \omega))^2 dt \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad (2)$$

Definition (Stochastic differential equation)

A process X_t satisfies the stochastic differential equation

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t \quad (3)$$

if it satisfies the integral equation

$$X_t = X_0 + \int_0^t b(s, X_s)ds + \int_0^t \sigma(s, X_s)dB_s \quad (4)$$

Theorem (Existence and uniqueness of SDEs (Part 1))

Let $T > 0$ and $b : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n, \sigma : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ be measurable functions satisfying

$$|b(t, x)| + |\sigma(t, x)| \leq C(1 + |x|) \quad x \in \mathbb{R}^n, t \in [0, T] \quad (5)$$

for some constant C , $|\sigma|^2 = \sum |\sigma_{ij}|^2$, and such that

$$|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq D|x - y| \quad x, y \in \mathbb{R}^n, t \in [0, T] \quad (6)$$

for some constant D . Let Z be a random variable which is independent of the σ -algebra F_∞ generated by $B_s, s \geq 0$, and such that

$$\mathbb{E}[|Z|^2] < \infty. \quad (7)$$

Theorem (Existence and uniqueness of SDEs (Part 2))

Then the stochastic differential equation

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \quad 0 \leq t \leq T, X_0 = Z \quad (8)$$

has unique continuous (in t) solution X_t with the property that X_t is adapted to the filtration $\mathcal{F}_t^Z$ generated by Z and $B_s, s \leq t$ and

$$\mathbb{E} \left[\int_0^T |X_t|^2 dt \right] < \infty. \quad (9)$$

Definition (Time-homogenous Itô diffusion)

A (time-homogenous) Itô diffusion is a stochastic process $X_t(\omega) = X(t, \omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R}^n$ satisfying a stochastic differential equation of the form

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t, \quad t \geq s; X_s = x$$

where B_t is m -dimensional Brownian motion and $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ satisfy

$$|b(x) - b(y)| + |\sigma(x) - \sigma(y)| \leq D|x - y|; \quad x, y \in \mathbb{R}^n$$

We call b the drift coefficient and σ the diffusion coefficient.

Theorem (Itô's lemma)

Let

$$dX(t) = udt + vB(t)$$

be an n -dimensional Itô process. Let $f(t, x)$ be a C^2 map from $[0, \infty) \times \mathbb{R}^n$ into $\mathbb{R}$. Then the process

$$Y(t, \omega) = f(t, X(t))$$

is again an Itô process, given by

$$dY = \frac{\partial f}{\partial t}(t, X)dt + \sum_i \frac{\partial f}{\partial x_i}(t, X)dX_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} dX_i dX_j$$

where $dB_i dB_j = \delta_{i,j} dt$, $dB_i dt = dt dB_i = dt dt = 0$.

Definition (Generator)

Let $\{X_t\}$ be an Itô diffusion in $\mathbb{R}^n$. The generator $\mathcal{L}$ of X_t is defined by

$$\mathcal{L}f(x) = \lim_{t \downarrow 0} \frac{\mathbb{E}^x[f(X_t)] - f(x)}{t}; \quad x \in \mathbb{R}^n$$

for the set of functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ where the limit exists uniformly.

Theorem (Diffusion generator)

The generator of the diffusion process is given by

$$\mathcal{L}_s f = b \cdot \nabla f + \frac{1}{2} \text{tr}(\sigma \sigma^T \nabla^2 f)$$

which in the case where σ is a constant simplifies to

$$\mathcal{L}_s f = b \cdot \nabla f + \frac{1}{2} \sigma^2 \Delta f$$

Here f is any C^2 function.

Proof

The Itô formula gives

$$df(X_t) = \sum_i b_i \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\sigma \sigma^T)_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} dt + M_t$$

where M_t is some martingale term. We rewrite this as

$$f(X_t) = f(x) + \int_0^t \mathcal{L}f(X_s) ds + M_t$$

and we now wish to show that $\mathcal{L}f$ is the generator.

Proof contd.

Taking expectations we have

$$T_t f(x) = f(x) + \int_0^t \mathbb{E}_x[\mathcal{L}f(X_s)] ds + \mathbb{E}[M_t]$$

so that

$$T_t f(x) - f(x) = \int_0^t T_s(\mathcal{L}f)(x) ds$$

which gives us that

$$\frac{d}{dt} T_t f(x)|_{t=0} = \mathcal{L}f(x)$$

and so it is clear that $\|\frac{d}{dt} T_t f - \mathcal{L}f\|$ as $t \rightarrow 0$.



Generative Models and Diffusion

Definition (Generative modelling)

Generative modelling is the process of training a model to transform a sample from a simple distribution p_{simple} to a data distribution p_{data} .

Definition (Dataset)

A dataset is a finite number of samples $z_1, \dots, z_N \sim p_{\text{data}}$.

Definition (Probability path)

A probability path is a time-dependent probability $\{p_t\}_{0 \leq t \leq 1}$.

Definition (Diffusion model)

A diffusion model is given by:

$$dX_t = b_t^\theta(X_t)dt + \sigma_t dW_t$$

$$X_0 \sim p_{\text{simple}}$$

where $b_t^\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the parametrized drift, $\sigma_t : \mathbb{R}^d \rightarrow \mathbb{R}$ is our diffusion coefficient at time t , and p_{simple} is our initial data distribution.

Aim

The goal is to find coefficients b_t^θ and σ_t such that $X_1 \sim p_{\text{data}}$.

Algorithm

Theorem

M_t^f is a martingale where

$$M_t^f = f(X_t) - f(X_0) - \int_0^t \mathcal{L}_s f(X_s) ds \quad (10)$$

so that $\mathbb{E}[M_t^f] = 0$. Here $f \in C^2$ is an arbitrary test function. $\mathcal{L}_s$ is the generator of the diffusion process.

Proof.

This follows from the proof of the diffusion generator. □

Naive residual

$$\tilde{R}_{t_k}^{f,i} = f(X_{t_{k+1}}^i) - f(X_{t_k}^i) - \int_{t_k}^{t_{k+1}} \mathcal{L}_s f(X_s^i) ds$$

Naive loss function

$$L_1 = \frac{1}{T|\mathcal{F}|} \sum_{k=1}^T \sum_{f \in \mathcal{F}} \left(\frac{1}{N} \sum_{i=1}^N \tilde{R}^{f,i} \right)^2 \quad (11)$$

Monte-Carlo loss function

$$L_2 = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \left(\frac{1}{N} \sum_{i=1}^N \tilde{R}^{f,i} \right)^2 \quad (12)$$

Weak residual

$$R_{t_k}^{f,i} = f(Y_{t_{k+1}}^i) - f(X_{t_k}^i) - \int_{t_k}^{t_{k+1}} \mathcal{L}_s f(X_s^i) ds \quad (13)$$

where $Y_{t_{k+1}} \sim p_{t_{k+1}}(\cdot|z)$ and $X_{t_k} \sim p_{t_k}(\cdot|z)$.

Weak loss

$$\tilde{L}_3 = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \left(\frac{1}{N} \sum_{i=1}^N R_{t_k}^{f,i} \right)^2 \quad (14)$$

Theorem

Minimizing equation 14 implies weak generator learning. Explicitly, if for each $f \in \mathcal{F}$ we have that $\mathbb{E}[R_t^f] = 0$, then

$$\mathbb{E}[\mathcal{L}f(X_t)] = \mathbb{E}[\mathcal{L}^\theta f(X_t)]$$

for each $f \in \mathcal{F}$.

Proof.

Using Thm. 16, we find that

$$\begin{aligned}\mathbb{E}[R_t] &= \mathbb{E}[f(Y_{t+\Delta t})] - \mathbb{E}[f(X_t)] - \int_t^{t+\Delta t} \mathbb{E}[\mathcal{L}^\theta f(X_s)] ds \\ &= \mathbb{E}[f(Y_0)] + \int_0^{t+\Delta t} \mathbb{E}[\mathcal{L}f(Y_s)] ds - \mathbb{E}[f(X_0)] - \int_0^t \mathbb{E}[\mathcal{L}f(X_s)] ds \\ &\quad - \int_t^{t+\Delta t} \mathbb{E}[\mathcal{L}^\theta f(X_s)] ds \\ &= \int_t^{t+\Delta t} \mathbb{E}[\mathcal{L}f(X_s) - \mathcal{L}^\theta f(X_s)] ds \\ &\approx \Delta t \mathbb{E}[\mathcal{L}f(X_t) - \mathcal{L}^\theta f(X_t)]\end{aligned}$$

Here we use Fubini's theorem, since the generator is integrable, and we use that X_s and Y_s follow the same distribution to get cancellations. $\square$

Error estimation

If we now write $\epsilon = \mathbb{E}[\mathcal{L}f(X_t) - \mathcal{L}^\theta f(X_t)]$ as our error, then

$$\min_{\theta} \left(\frac{1}{N} \sum_{i=1}^n R_{\theta}^{f,i} \right)^2 \sim \Delta t^2 \epsilon^2 \quad (15)$$

Decomposition

$$\mathbb{E}[(R_\theta^f)^2] = (\mathbb{E}[R_\theta^f])^2 + \text{Var}(R_\theta^f),$$

Martingale error

$$L_{ME} = \frac{1}{|\mathcal{F}|N} \sum_{f \in \mathcal{F}} \sum_{i=1}^N (R^{f,i})^2. \quad (16)$$

Algorithm Martingale Matching

Require: Neural net b_t^θ , diffusion coefficient σ , time-step Δt , dataset of samples $z \sim p_{data}$, conditional probability path $p_t(\cdot|z)$

for each mini-batch of data **do**

 Sample datum z from dataset.

 Sample $t \sim U[0, 1 - \Delta t]$.

 Set $x_t \sim p_t(\cdot|z)$, $x_{t+\Delta t} \sim p_{t+\Delta t}(\cdot|z)$

 Compute loss $\mathcal{L}_{ME}^\theta$

 Update the model parameter θ via gradient descent on $\mathcal{L}_{ME}^\theta$

end for

Definition (Random Fourier features)

A random Fourier feature is a function of the type

$$f_{\omega_1}(x) = \cos(w^\top x) \quad f_{\omega_2}(x) = \sin(w^\top x)$$

where $x \in \mathbb{R}^d$ and $\omega \sim N(0, \sigma^2 I_d)$.

Definition (Polynomial test functions)

Suppose our dataset has dimension n , and we want to consider polynomials of degree d . The the set of polynomial test functions is

$$\{f(x) = x_i^k x_j^k | i, j \in \{0, \dots, n\}, k \in \{0, \dots, d\}\}.$$

Definition (Gaussian probability path)

Let $\alpha_t = \alpha(t) \in C^1([0, 1])$ and $\beta_t = \beta(t) \in C^1([0, 1])$ be noise schedulers, that is, monotonic functions with $\alpha_0 = \beta_1 = 0$ and $\alpha_1 = \beta_0 = 1$. We then define the Gaussian conditional probability path to be

$$p_t(\cdot|z) = N(\alpha_t z, \beta_t^2 I_d).$$

Then this probability path fulfills the boundary conditions

$$p_0(\cdot|z) = N(\alpha_0 z, \beta_0^2 I_d) = N(0, I_d) \quad \text{and} \quad p_1(\cdot|z) = N(\alpha_1 z, \beta_1^2 I_d) = N(z, 0)$$

Algorithm Euler-Maruyama

Require: Neural net b_t^θ , number of steps n , diffusion coefficient σ_t

Set $h = \frac{1}{n}$

Set $t = 0$

Sample $X_0 \sim p_{\text{simple}}$

for $i = 1, \dots, n$ **do**

 Sample $\epsilon \sim N(0, I)$

$X_{t+h} = X_t + hb_t^\theta(X_t) + \sigma_t \sqrt{h} \epsilon$

 Update $t \leftarrow t + h$

end for

return X_1

Results

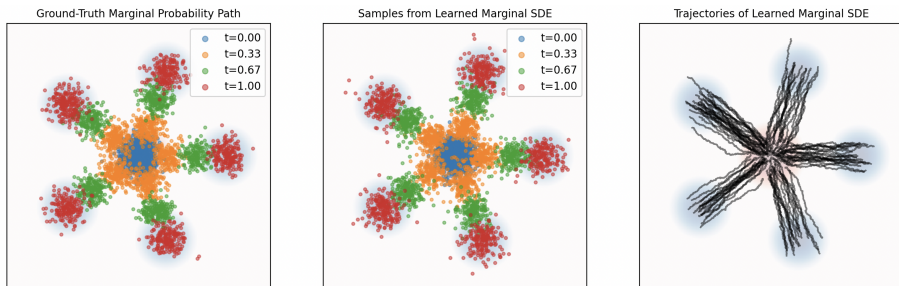


Figure: Learning multiple Gaussians

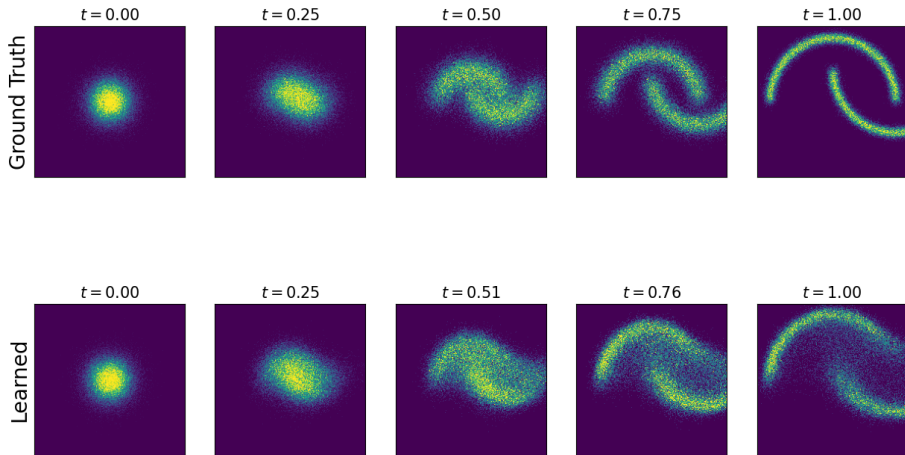


Figure: Learning two moons with Fourier test functions

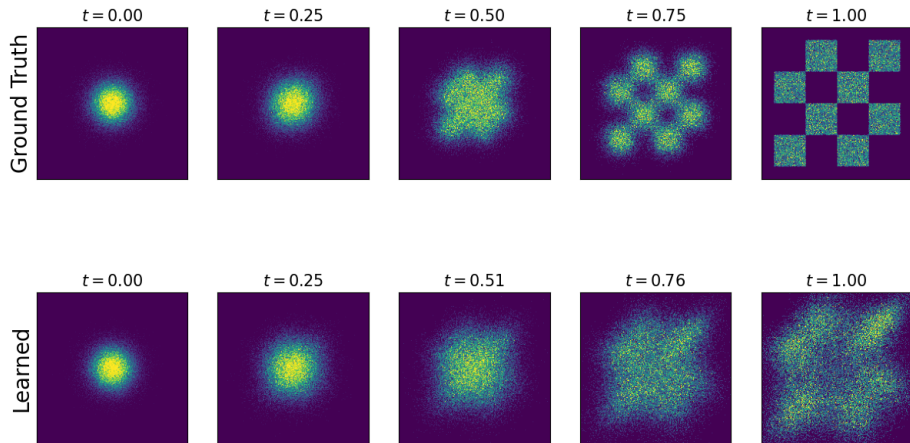


Figure: Learning a checkerboard with Fourier test functions

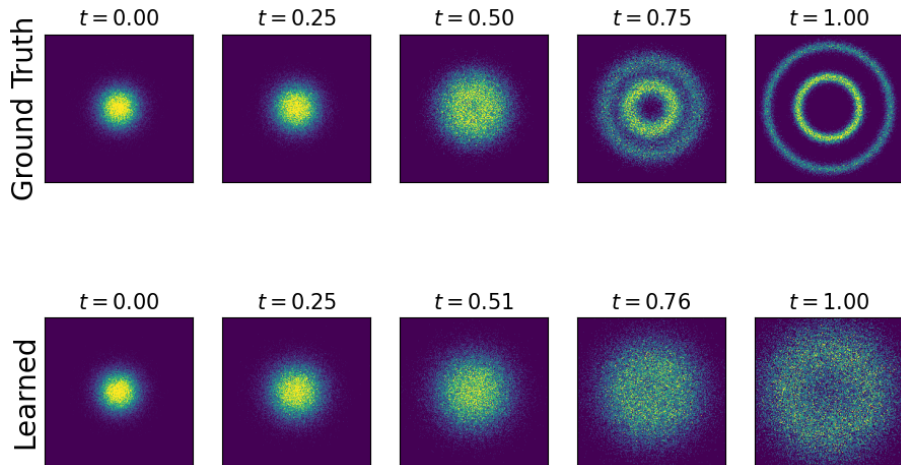


Figure: Learning circles with Fourier test functions

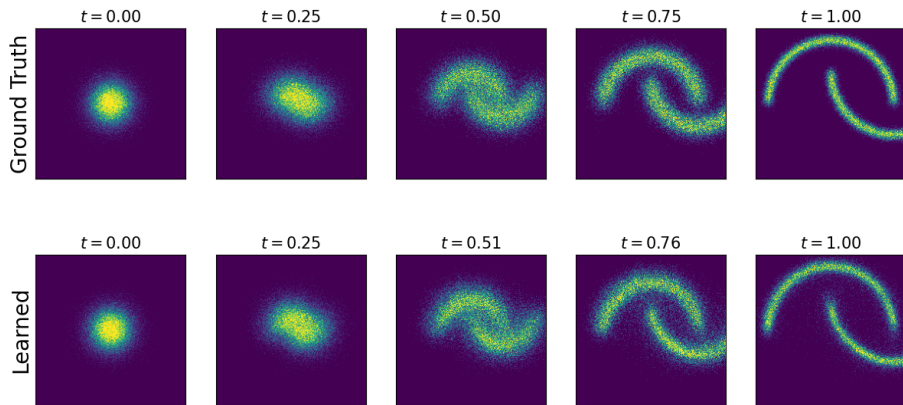


Figure: Learning two moons with polynomial test functions

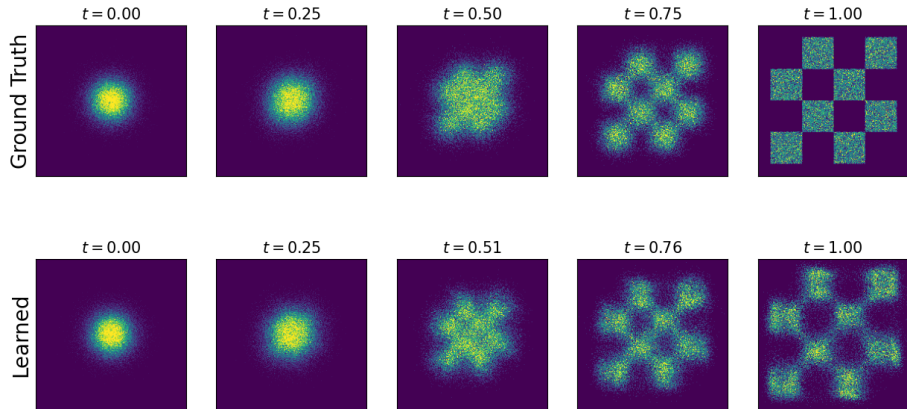


Figure: Learning a checkerboard with polynomial test functions

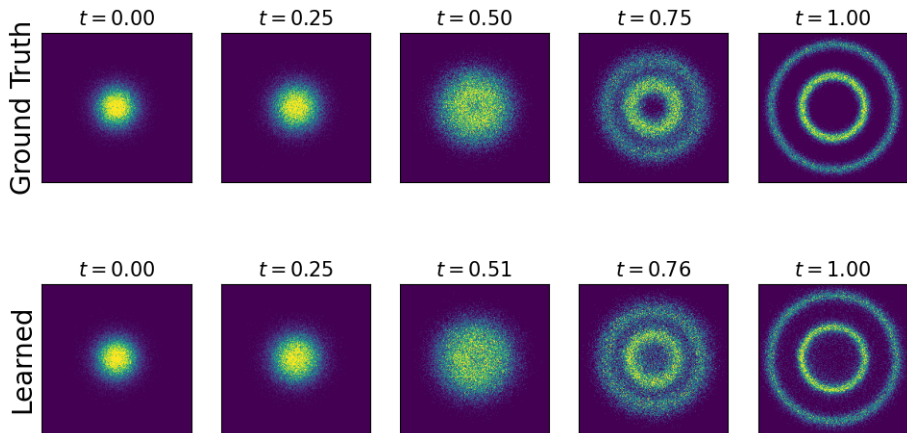


Figure: Learning circles with polynomial test functions

Conclusion

- We get excellent results for lower dimensional data
- Polynomial test functions expensive in higher dimensions

Extensions

- Combining test functions
- Learning the test functions
- More thorough analysis of what test functions learn what

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